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Summary

Researcher in computational astronomy specializing in planet formation simulations, studying the formation of planets from planetesimals with large-scale, self-consistent *N*-body simulations. PI of three consecutive "Fugaku" Young Researcher projects and a Miyabi "Large-scale HPC Challenge" project; chief developer of GPLUM-GPU, a high-performance GPU-accelerated *N*-body simulation code for planet formation.

Research Fields: Natural Sciences / Space and Planetary Science / Planetary Formation; Information and Communication / High-Performance Computing / Parallel Computing

Education

Apr 2024 – present	**Ph.D. Program, Department of Planetology, Graduate School of Science, Kobe University** Computational Planetology Group · Advisor: Prof. Junichiro Makino
Apr 2022 – Mar 2024	**M.Sc. in Planetology, Graduate School of Science, Kobe University**
Apr 2020 – Mar 2022	**B.Sc. in Planetology, Faculty of Science, Kobe University** Admitted by transfer in the third year
Apr 2017 – Mar 2020	**Faculty of Environment and Information Studies, Keio University** Major in computer science · Withdrew to transfer to Kobe University
Jun 2016	**Glebe Collegiate Institute, Ottawa, Ontario, Canada**

Positions, Fellowships & Software

Apr 2025 – Mar 2027	**JSPS Research Fellow for Young Scientists (DC2)** Japan Society for the Promotion of Science · Project: "Exploring the diversity and origins of planetary systems through large-scale numerical simulations" (KAKENHI 25KJ1814)
Present	**Chief Developer, GPLUM-GPU** A high-performance GPU-accelerated *N*-body simulation code for planet formation (particle–particle particle–tree scheme) · github.com/GPLUM-GPUAdmin

Research Grants & Computational Allocations (as Principal Investigator)

FY2026	**"Large-scale HPC Challenge" Project, Miyabi Supercomputer System, Joint Center for Advanced High Performance Computing (JCAHPC)** — Investigation of a GPU computing framework for ultra-high-resolution planet formation simulations resolving individual planetesimals
Apr 2026 – Sep 2026	**FY2026 "Fugaku" Young Researcher Project** — Unified large-scale *N*-body simulation of planet formation from planetesimals considering disk wind and viscous diffusion
Apr 2025 – Mar 2026	**FY2025 "Fugaku" Young Researcher Project** — Unified large-scale *N*-body simulation of planet formation from planetesimals considering planetary migration
Apr 2024 – Mar 2025	**FY2024 "Fugaku" Young Researcher Project** — Large-scale *N*-body simulation of planet formation: The effect of planetesimal-driven migration on the planet formation process

Publications

REFEREED JOURNAL PAPERS (FIRST AUTHOR)

[1] **T. Jinno**, Y. Ishigaki, T. R. Saitoh, Y. Funato, J. Makino, **"GPLUM-GPU: A high-performance GPU-accelerated *N*-body simulation code for planet formation with a particle–particle particle–tree scheme"**, submitted to *Publications of the Astronomical Society of Japan*

[2] **T. Jinno**, T. R. Saitoh, Y. Funato, J. Makino, **"Self-consistent *N*-body simulation of planetesimal-driven migration. II. The effect of PDM on planet formation from a planetesimal disk"**, *Publications of the Astronomical Society of Japan*, 78(2), 764–780 (2026). doi:10.1093/pasj/psag017

[3] **T. Jinno**, T. R. Saitoh, Y. Funato, J. Makino, **"Self-consistent *N*-body simulation of planetesimal-driven migration. I. The trajectories of single planets in the uniform background"**, *Publications of the Astronomical Society of Japan*, 76(6), 1309–1322 (2024). doi:10.1093/pasj/psae090

[4] **T. Jinno**, T. R. Saitoh, Y. Ishigaki, J. Makino, **"*N*-body simulation of planetary formation through pebble accretion in a radially structured protoplanetary disk"**, *Publications of the Astronomical Society of Japan*, 75(5), 951–969 (2023). doi:10.1093/pasj/psad053

[5] **T. Jinno**, T. R. Saitoh, Y. Funato, J. Makino, **"Unified large-scale *N*-body simulation of planet formation from planetesimals considering planetary migration"** (in Japanese), *Planetary People (Yuseijin), The Japanese Society for Planetary Sciences*, 35(2), 54–65 (2026). doi:10.14909/yuseijin.35.2_54

REFEREED JOURNAL PAPERS (CO-AUTHOR)

[6] D. M. Hernandez, **T. Jinno**, J. Makino, **"Time-reversible timestepping in many-body planetary systems"**, submitted to *Monthly Notices of the Royal Astronomical Society*. arXiv:2610.22007

REFEREED CONFERENCE PROCEEDINGS (FIRST AUTHOR)

[7] **T. Jinno**, T. R. Saitoh, Y. Ishigaki, J. Makino, **"*N*-Body Simulations of Planet Formation within the Inner Dead-Zone Boundary of Protoplanetary Disk"**, in *Proceedings of the 34th IUPAP Conference on Computational Physics (CCP2023)*, Springer Proceedings in Physics, vol. 356, Springer, Singapore (2026). doi:10.1007/978-981-92-0844-9_30

NON-REFEREED

[8] M. Sekiguchi, R. Ishizaki, K. Shobuzako, **T. Jinno**, T. Yoneya, A. Yoshida, X. Yu, **"Report on the Planetary Science Young Researchers Workshop 2024"** (in Japanese), *Yuseijin, The Japanese Society for Planetary Sciences*, 33(3) (2024). doi:10.14909/yuseijin.33.3_356

Conference Presentations

Presenting author on all items; co-authors as in the related papers.

INVITED TALKS

Apr 2025	**Global *N*-body Simulation of Planetary Formation: The Origins of Ice Giants** — International STEAM Symposium for Young Researchers 2025	Invited

INTERNATIONAL CONFERENCES

Aug 2026	**Development of a GPU-accelerated large-scale *N*-body simulation code for planet formation** — Formation of Stars and Planets in the Evolving Galaxy, Okinawa	Poster
May 2026	**Planet Formation in the Era of 100 Million-Planetesimal *N*-body Simulations** — JpGU-AGU Joint Meeting 2026, Chiba	Oral
Dec 2025	**The effect of planetesimal-driven migration on planet formation from planetesimals** — Int. Conf. on Exoplanets and Planet Formation 2025, Shanghai	Poster
Sep 2025	**Global *N*-body simulation of planetary formation: The origins of ice giants** — EPSC-DPS Joint Meeting 2025, Helsinki	Poster
Jul 2024	**Global *N*-body simulation of planet formation: The effects of planetesimal-driven migration on planetary formation process** — International HPC Summer School 2024, Kobe	Poster
Aug 2023	***N*-body simulation of planetary formation through pebble accretion in a radially structured protoplanetary disk** — CCP2023 (34th IUPAP Conf. on Computational Physics), Kobe	Poster

DOMESTIC CONFERENCES (JAPAN)

16 contributed presentations (10 oral, 6 poster), Mar 2023 – Oct 2026, as presenting author at:

- Astronomical Society of Japan, Annual Meetings (Spring 2023, Autumn 2023, Spring 2024, Spring 2025, Autumn 2026) — 5 oral
- Japan Geoscience Union Meetings (2023, 2024, 2025) — 2 oral, 1 poster
- The Japanese Society for Planetary Sciences, Autumn Meeting 2024 — 1 oral
- Planet Formation Workshop 2025 — 1 oral
- Summer School on Astronomy and Astrophysics (53rd, 2023; 54th, 2024) — 1 oral, 1 poster
- HPCI System Research Results Presentations on "Fugaku" (12th, 2025; 13th, 2026) — 2 poster
- Research Workshops of the Program for Promoting Research on the Supercomputer Fugaku (3rd, 2024; 5th, 2026) — 2 poster

Awards & Honors

2026	**Outstanding Presentation Award, 15th Science Frontier Research Presentation (FY2026)**
May 2026	**Student Outstanding Presentation Award, Space and Planetary Sciences Section, JpGU-AGU Joint Meeting 2026**
2025	**Outstanding Presentation Award, 14th Science Frontier Research Presentation (FY2025)**
Sep 2024	**Best Presentation Award, The Japanese Society for Planetary Sciences 2024**
Jul 2024	**Selected as Japan Representative, International HPC Summer School 2024**
2024	**Outstanding Presentation Award, 13th Science Frontier Research Presentation (FY2024)**
Nov 2023	**Outstanding Presentation Award, 12th Science Frontier Research Presentation (FY2023)**
Aug 2023	**1st Place Oral Award, ISM/Star & Planet Formation Session, 53rd Summer School on Astronomy and Astrophysics**

Skills

- **Simulation & HPC:** Large-scale parallel *N*-body simulation of planet formation; GPU-accelerated code development (chief developer of GPLUM-GPU); production runs on the supercomputer "Fugaku" and the GPU supercomputer Miyabi
- **Programming:** C/C++, CUDA, MPI/OpenMP, Python
- **Languages:** Japanese (native), English (fluent; TOEIC Listening & Reading 980; high-school education in Canada)
- **Certificates:** 2024 International Summer School on HPC Challenges in Computational Sciences, Certificate of Participation; Class 4 Amateur Radio Operator License (Japan)

Professional Memberships & Community Involvement

- **Member:** Astronomical Society of Japan; The Japanese Society for Planetary Sciences; Japan Geoscience Union
- **2024:** Organizing Committee Member, Planetary Science Young Researchers Workshop 2024 (The Japanese Society for Planetary Sciences)